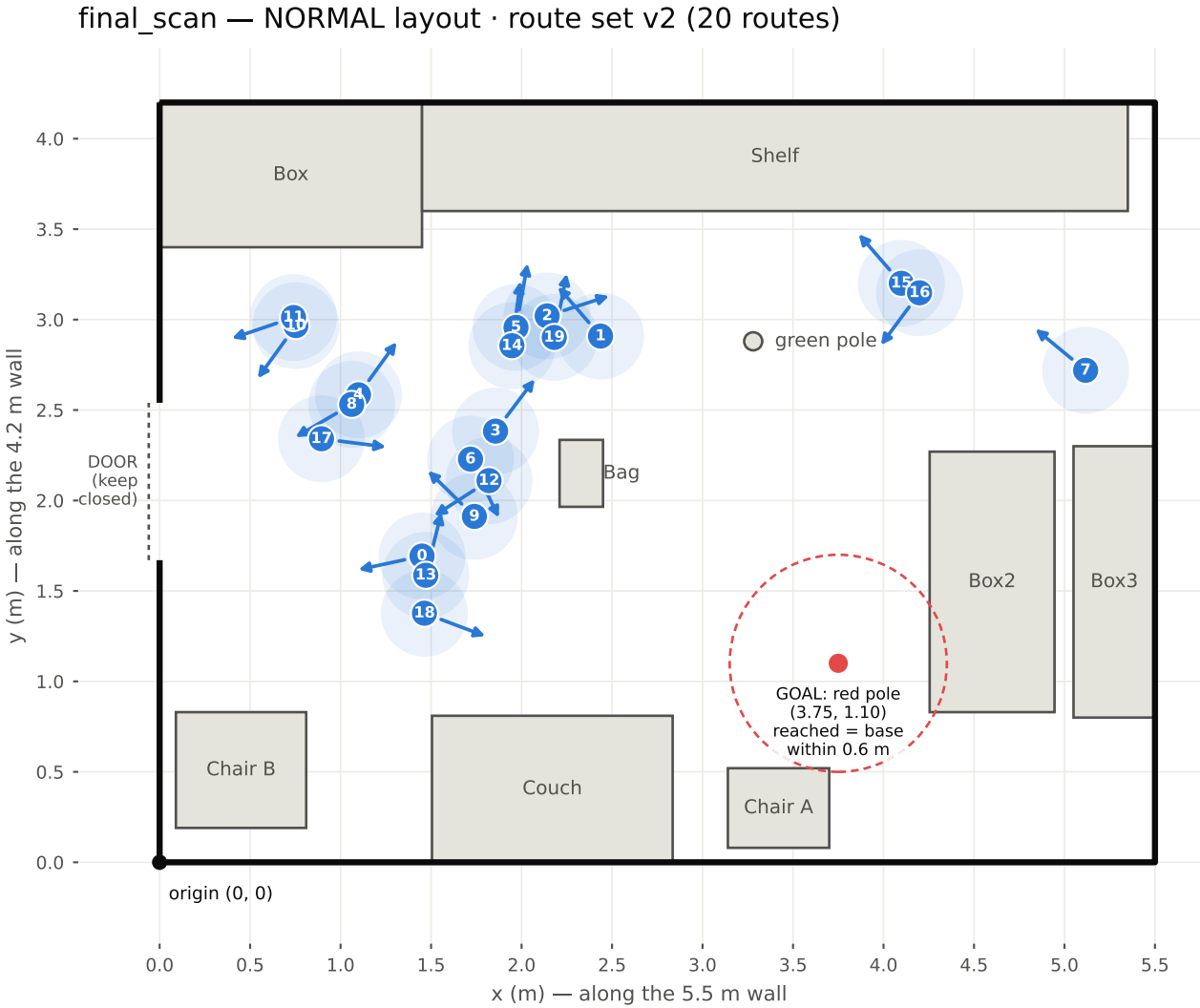
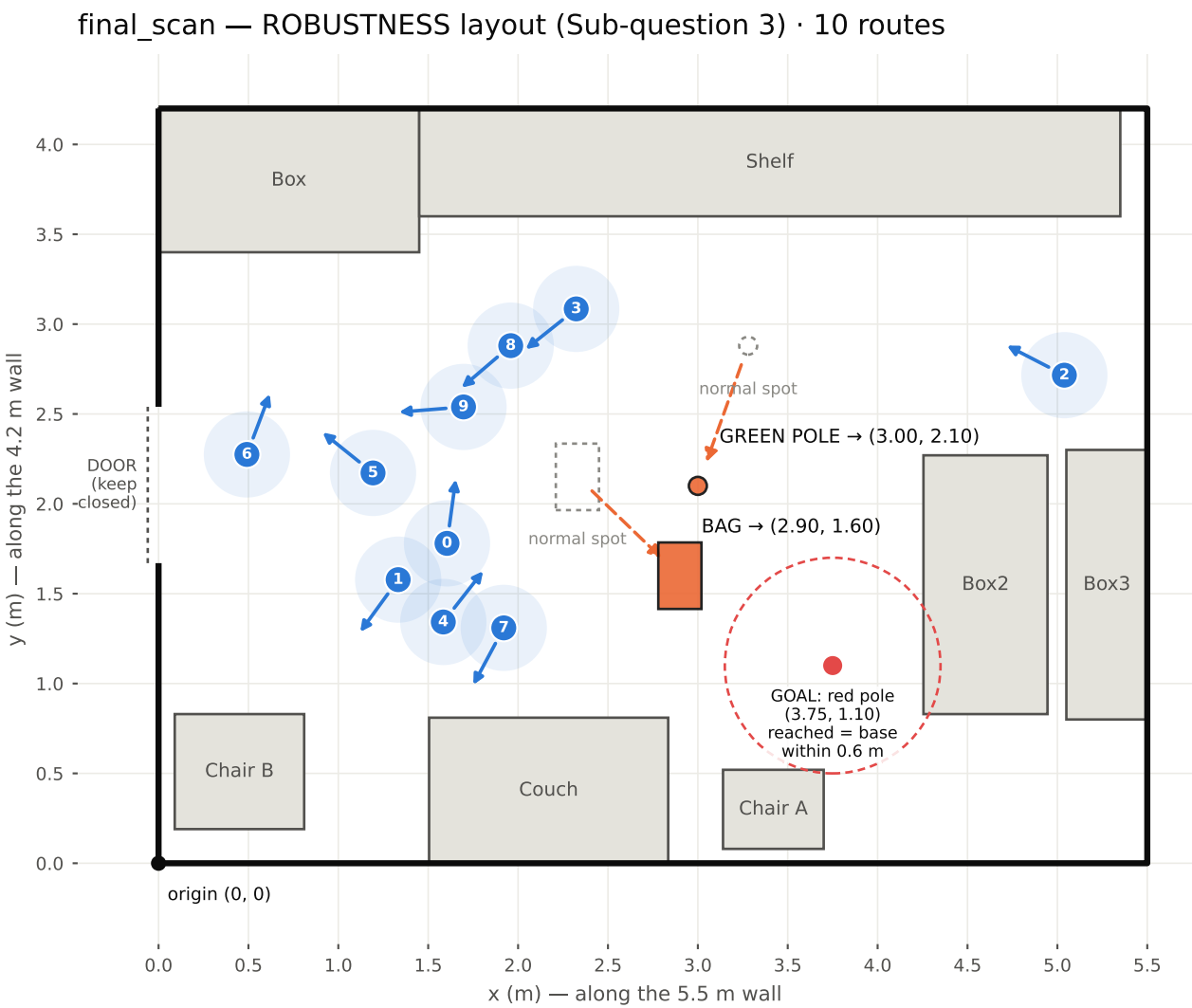


route	x (m)	y (m)	heading	l* (m)	done
r000	1.45	1.69	-168°	1.87	
r001	2.44	2.91	+131°	1.73	
r002	2.14	3.02	+18°	1.99	
r003	1.86	2.38	+53°	2.06	
r004	1.10	2.58	+54°	2.65	
r005	1.97	2.96	+80°	2.12	
r006	1.72	2.23	-64°	1.97	
r007	5.12	2.72	+141°	2.04	
r008	1.06	2.53	-150°	2.63	
r009	1.74	1.91	+136°	1.73	
r010	0.75	2.97	-126°	3.11	
r011	0.74	3.01	-161°	3.18	
r012	1.82	2.11	-147°	1.83	
r013	1.47	1.59	+76°	1.83	
r014	1.95	2.86	+82°	2.13	
r015	4.10	3.20	+131°	1.63	
r016	4.20	3.15	-126°	1.57	
r017	0.89	2.34	-7°	2.74	
r018	1.46	1.38	-21°	1.75	
r019	2.18	2.90	+79°	1.90	

Viewed from above. Numbered blue dot = centre of Pepper's base (route number); arrow = the way she faces (heading measured from +x, anticlockwise positive); pale disc = her footprint. Tape a cross on each dot and a 40 cm line along each arrow. Use with: eval.py --variant v2. Overlapping dots (0/13, 2/19, 4/8, 5/14, 10/11, 15/16) are starts a few cm apart, not a drawing error: tape those from the table's coordinates.



route	x (m)	y (m)	heading	l* (m)	done
r000	1.60	1.78	+82°	2.78	
r001	1.33	1.58	-126°	3.17	
r002	5.04	2.72	+153°	1.94	
r003	2.32	3.08	-141°	2.04	
r004	1.58	1.34	+52°	3.02	
r005	1.19	2.17	+141°	3.07	
r006	0.49	2.27	+69°	3.73	
r007	1.92	1.31	-118°	2.79	
r008	1.96	2.88	-139°	2.31	
r009	1.70	2.54	-175°	2.47	

Viewed from above. Numbered blue dot = centre of Pepper's base (route number); arrow = the way she faces (heading measured from +x, anticlockwise positive); pale disc = her footprint. Tape a cross on each dot and a 40 cm line along each arrow. Move ONLY the two orange pieces, tape-measuring their centres; everything else stays. Run after both normal passes. Use with: eval.py --variant perturbed. Put them back afterwards.